ChatGPT said:

A. Linux Fundamentals and File System

We will cover **Section A only**, one question at a time, using short, simple, interview-ready answers suitable for a Cloud/DevOps Engineer with 1+ year of experience.

A1. What is Linux? What is the difference between Linux kernel and Linux distribution?

Answer:

Linux is an open-source, Unix-like operating system used to manage hardware, memory, processes, files and applications.

- **Linux Kernel:** The core of the operating system. It manages CPU, memory, hardware, processes and system resources.
- **Linux Distribution:** A complete operating system built around the Linux kernel, along with system utilities, package managers and other software.

Examples of Linux distributions:

- Amazon Linux
- Ubuntu
- Red Hat Enterprise Linux (RHEL)
- CentOS
- Debian

Interview example:

If I launch an EC2 instance using Amazon Linux 2023, I am using a Linux distribution that has the Linux kernel along with system tools and package management utilities.

A2. What is the difference between Linux and Unix?

Answer:

Linux	Unix
Open-source	Mostly proprietary
Generally free	Often requires licensing
Developed by the open-source community	Developed by companies and organizations
Examples: Ubuntu, Amazon Linux	Examples: AIX, HP-UX, Solaris

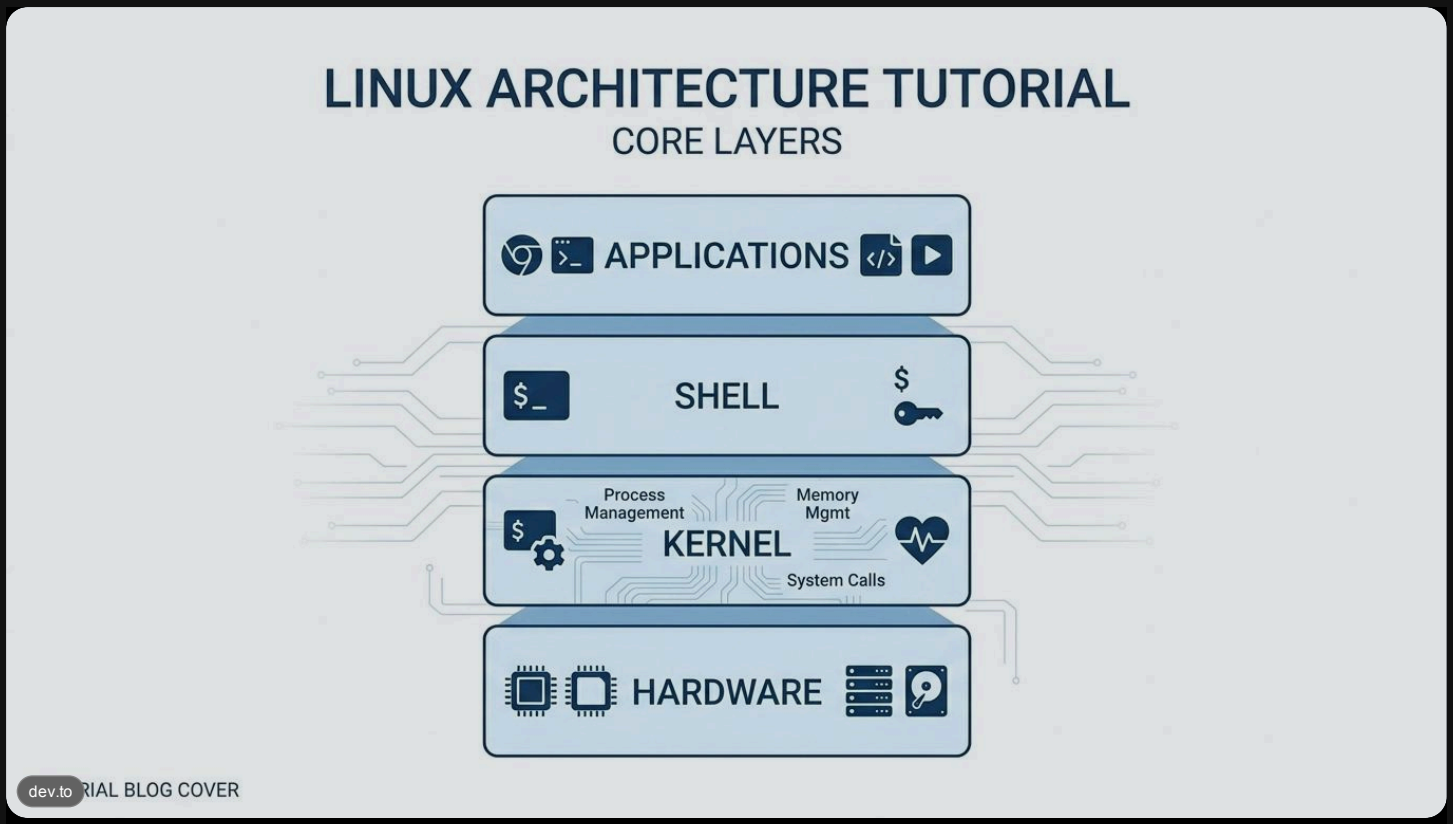
In simple words: Linux is Unix-like, meaning it follows many Unix concepts and design principles, but it is not originally derived from Unix source code.

A3. Explain Linux architecture.

Answer:

Linux architecture consists of four main layers:

1. **Hardware:** Physical components such as CPU, RAM, disk and network devices.
2. **Kernel:** Manages hardware, memory, processes and system resources.
3. **Shell:** Acts as an interface between the user and the kernel. Examples: Bash and Zsh.
4. **Applications:** Programs and utilities that users execute.



Example: When you execute `ls`, the shell processes the command, requests the necessary operations through the operating system, and displays the output.

A4. Explain the Linux file system hierarchy.

Answer:

Linux uses a hierarchical file system that starts from the root directory `/`.

Important directories:

Directory	Purpose
`/`	Root directory; starting point of the file system
`/etc`	Configuration files
`/var`	Logs and frequently changing data
`/tmp`	Temporary files
`/home`	Users' personal directories
`/root`	Root user's home directory
`/opt`	Optional third-party applications
`/usr`	User applications, binaries and libraries
`/bin`	Essential user commands
`/sbin`	System administration commands
`/boot`	Bootloader and kernel files
`/dev`	Device files
`/proc`	Process and kernel information
`/mnt`	Temporary mount points
`/media`	Removable media mount points

DevOps example: If an application fails, I generally check `/var/log` for logs and `/etc` for configuration files.

A5. What is the difference between absolute path and relative path?

Answer:

- **Absolute path:** Complete path starting from the root directory `/`.
- **Relative path:** Path specified relative to the current working directory.

Example:

```bash
# Absolute path
cd /var/log

# Relative path
cd logs
```

If my current directory is `/var`, then `cd logs` takes me to `/var/log`.

A6. What is the difference between hard link and soft link?

Answer:

Hard Link	Soft Link
Points to the same inode	Points to the original file path
Shares the same file data	Has a separate inode
Works even if the original filename is deleted	Can break if the original file is deleted
Generally cannot link directories	Can link directories

Commands:

```bash
# Create hard link
ln file.txt hardlink.txt

# Create soft link
ln -s file.txt softlink.txt
```

Simple example: A hard link is another filename for the same underlying file data, whereas a soft link is like a shortcut pointing to another file.

A7. What are hidden files in Linux?

Answer:

Files or directories whose names start with a dot (`.`) are hidden files in Linux.

They are generally used for configuration and application settings.

Examples:

- `.bashrc`
- `.bash_profile`
- `.ssh`
- `.gitconfig`

Command to view hidden files:

```bash
```

```
ls -la
```

A8. What are environment variables in Linux?

Answer:

Environment variables are key-value pairs used to store configuration information that can be accessed by processes and applications.

Examples:

- `PATH` – Directories where Linux searches for executable commands.
- `HOME` – Current user's home directory.
- `USER` – Current username.
- `SHELL` – Current user's default shell.

Commands:

Bash

```
# View environment variables
env

# View a specific variable
echo $HOME

# Create an environment variable
export APP_ENV=production
```

A9. Explain stdin, stdout and stderr.

Answer:

Linux uses three standard streams:

Stream	Number	Meaning
stdin	0	Standard input
stdout	1	Standard output
stderr	2	Standard error

Example:

Bash

```
# Save normal output
ls > output.txt

# Save errors
ls /invalid 2> error.txt

# Save both output and errors
ls /home /invalid > output.txt 2>&1
```

In simple words:

- stdin: Input given to a command.
- stdout: Normal output.
- stderr: Error messages.

A10. What is the difference between `>` and `>>`?

Answer:

Both operators redirect output into a file.

- `>` overwrites existing file content.
- `>>` appends output to existing file content.

Example:

```bash
echo "Hello" > file.txt
```

Overwrites the file.

```bash
echo "World" >> file.txt
```

Adds content at the end without deleting existing content.

A11. What is a pipe in Linux?

Answer:

A pipe (`|`) passes the output of one command as input to another command.

It helps combine multiple commands to perform operations efficiently.

Example:

```bash
ps aux | grep nginx
```

Here:

- `ps aux` displays running processes.
- `grep nginx` filters processes containing nginx.

DevOps example: We commonly use pipes to filter logs, identify processes and troubleshoot production issues.

Section A – Commands to practise

```bash
pwd
ls -la
cd
tree
which
whereis
find
locate
ln
echo
env
printenv
export
```

Section A completed

My recommendation: Don't just memorize these answers. Practise the commands on your Amazon Linux EC2 instance and explain each concept aloud as if you are sitting in an interview.

Next, we will cover **B. File and Directory Management**, following the exact same question-by-question format.